

Tyler Jay Yang

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EDUCATION AND HONORS

Carnegie Mellon University, School of Computer Science Pittsburgh, PA August 2024 – May 2027
Bachelor of Science, Computer Science, **GPA: 4.0** | Dean's List
Honors: 2x American Invitational Mathematics Examination (AIME), National Merit Scholarship Recipient

Relevant Coursework: Linear Algebra, Vector Calculus, Probability & Statistics, Data Structures & Algorithms, Discrete Math, Theoretical CS, Computer Systems, Deep Learning, Computational Biology, Quantum Computation

EXPERIENCE

Technical Advisor Intern August 2025 – Present
Scale AI

- Trained models for complex reasoning tasks and identified model failure modes
- Collaborated with researchers and interns to advance AI capabilities and safety

Computational Chemistry Research Assistant February 2025 – August 2025
Mechanical and Artificial Intelligence Lab, Carnegie Mellon University

- Engineered synthetic data generation pipeline to fine-tune VLMs for document extraction tasks, increased efficiency **4x** using Python and Ray actors for multithreaded concurrency, and embedded into reaction extraction pipeline with classical natural-language processing tools
- Developed transformer encoder and decoder from scratch in *PyTorch* for solubility estimation and retrosynthetic prediction from SMILES strings

Machine Learning Intern December 2022 – August 2023
Hillman Cancer Center, University of Pittsburgh Medical Center

- Created *ImageJ Java scripts* to preprocess fluorescence microscopy images to train *MATLAB* image segmentation algorithm to isolate thousands of asymmetrically dividing T cell clusters
- Achieved **90% accuracy**, outperforming Sartorius commercial software
- Published (peer-reviewed) doi.org/10.1016/j.semcd.2023.08.001

PROJECTS

Mitigating Algorithmic Bias in Skin Cancer Detection

- Added support for per-epoch adaptive resampling module in *Keras* data loaders
- Integrated with *Tensorflow* training pipeline and automated hyperparameter search using *Sklearn* and *Comet* to improve model accuracy by **31%**
- Optimized image I/O performance via *HDF5* storage and *Pandas* preprocessing

Quantum Computing Applications

- Designed custom libraries in C for qubit operations optimized with *Valgrind* to support **100x** more qubits for simulating quantum algorithms
- Partnered with UNICC, IDQuantique to co-design the innovation challenge for *Future Leaders in Quantum Hackathon 2025*, connecting over 600 participants across 30 countries with quantum industry partners

Sequence Analysis Toolkit

- Developed pathfinding and optimization algorithms for genome assembly and sequence alignment
- Built Hidden Markov Models using *NumPy* and implemented the Viterbi algorithm for genome reconstruction
- Performed Principal Component Analysis (PCA) in R with *ggplot* to classify SARS-CoV-2 variants

ACTIVITIES

Teaching Assistant, CMU Math Department, 21-259 Calculus in Three Dimensions August 2025 – Present
Board Member, Carnegie Mellon Computational Biology Undergraduate Society January 2025 – Present
Club Officer, Carnegie Mellon Quantum Computing Club August 2024 – Present
Futsal Club October 2024 – Present
Lab Supervisor, University of Memphis Satellite Campus August 2022 – May 2023

SKILLS

Languages: Python, C, C++, JavaScript, Java, Julia, R, MATLAB

Machine Learning and Big Data: Tensorflow/Keras, PyTorch, Sklearn, Pandas, NumPy, Matplotlib, Flask, SQL

Other: Git, GitHub, LaTeX, BASH scripting, Blender, Arduino, SolidWorks CAD, 3-D Printing